RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. in Information Technology
First Year - Semester I Examination – February 2024

ICT 1402 – PRINCIPLES OF PROGRAM DESIGN AND PROGRAMMING (PRACTICAL)

Time: Two (02) hours

- This paper has three (03) questions in four (04) pages.
- Answer all questions.
- The total mark allotted for this paper is 80.
- Note that questions appear on both sides of the paper.
- If a page is not printed, please inform the supervisor immediately.
- Create a folder on the desktop and name it with the examination index number.
- Save all the source codes and executable files inside the folder created on the desktop.

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1. Write a C program to automate the distance matrix shown in Figure 1, following the provided instructions. (40 Marks)

	AMP	ANU	AVS	BAD
AMB	0			
ANU	154	0		
AVS	42	195	0	
BAD	176	270	171	0

Figure 1: Distance matrix (AMB, ANU, AVS, and BAD are cities).

Assume that the route numbers between cities are determined based on the ROW-COLUMN basis. Accordingly, the route number for ANU-AMP is 10, and for AVS-AMP, it is 21 (note that the index for the first row and column is considered as 0). Consequently, it is possible to derive six different route numbers, as shown in Table 1, for six different routes based on the distance matrix provided in Figure 1.

Table 1: Route Numbers

Route	Route Number
ANU-AMP	10
AVS-AMP	20
BAD-AMP	30
AVS-ANU	21
BAD-ANU	31
BAD-AVS	32

i. Create a C source file and name it distance_calc.c.

ii. Create a two-dimensional array named distance_mat to store the distance values as shown in the distance matrix provided in Figure 1.

iii. Extend the program to enable users to query the distance of a specific route by entering a route number. The program should continuously prompt users for route numbers until they decide to exit by entering 0. The expected output of the program is depicted in Figure 2 for further clarification.

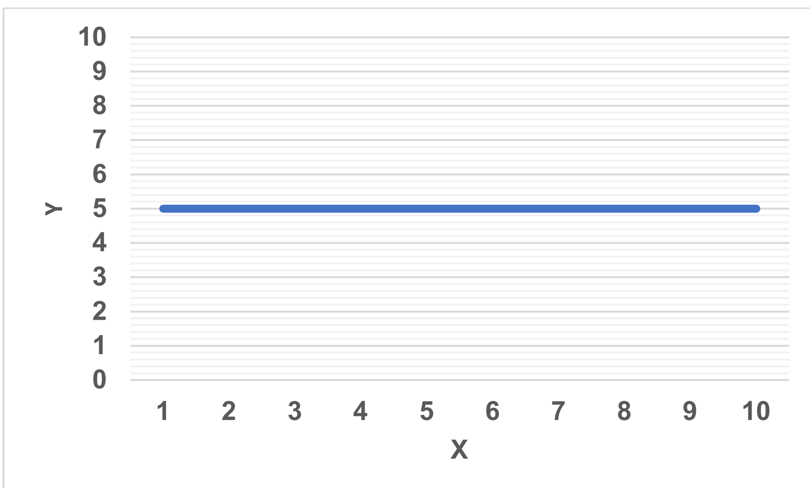
```
Enter Route Number : 10
Distance of route nuber 10 is 154 km

Enter Route Number : 20
Distance of route nuber 20 is 42 km

Enter Route Number : 30
Distance of route nuber 30 is 176 km

Enter Route Number : 21
Distance of route nuber 21 is 195 km

Enter Route Number : 31
Distance of route nuber 31 is 270 km

Enter Route Number : 32
Distance of route nuber 32 is 171 km

Enter Route Number : 11
Invaid route number

Enter Route Number : 0
Good Bye

Press Enter to return to Quincy...
```

Figure 2: Expected output of the distance_calc.c program.

iv. It is mandatory to use a switch statement to handle messages according to the selected route.

v. It is also mandatory to read the distance_mat array created in Question 1 (ii) to display the distance.

2. Figure 3 depicts the graph of Y=5. Write a C program to check whether a given XY coordinate is above the line, below the line, or on the line. For example, if the user inputs coordinate as (2, 3), the program should print the message "Below the line". If the user inputs coordinate as (5, 8), the program should print "Above the line". And if the user inputs coordinate as (4, 5), the program should print "On the line"." Save source file as linetest.c. (12 Marks)



X	Y
1	5
2	5
3	5
4	5
5	5
6	5
7	5
8	5
9	5
10	5

Figure 3: Y=5 line.

3. Write a C program that calculates the total area of a series of rectangular rooms, based on the given instructions. (28 Marks)

i. Create a C header file and name it roomanalysis.h.

ii. Create a function in the header file to return the area of a room using the following function prototype:

```
double calculateRectArea (double length, double breadth);
```

iii. Create a C source file and name it myrooms.c.

iv. Write a C program in myrooms.c that functions as a basic room analysis tool. In this program, the user must first input the total number of rooms in the building.

Based on the entered number of rooms, the program will iteratively prompt the user to enter the length and breadth for each room.

Subsequently, the program will call the room area calculation function from roomanalysis.h to calculate the area of the room based on the entered length and breadth.

Additionally, the program will print the area of each room and the total area of the building at the end of the execution.

The expected output of this proposed program is given in Figure 4 for further clarification.

```
Enter # of rooms : 2

Enter Length of room # 1: 10.5
Enter Breadth of room # 1: 12
Room no 1 area is 126.00 square meters

Enter Length of room # 2: 13
Enter Breadth of room # 2: 14
Room no 2 area is 182.00 square meters

Total area : 308.00 square meters

Press Enter to return to Quincy...
```

Figure 4: Expected output of the myrooms.c program.

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